

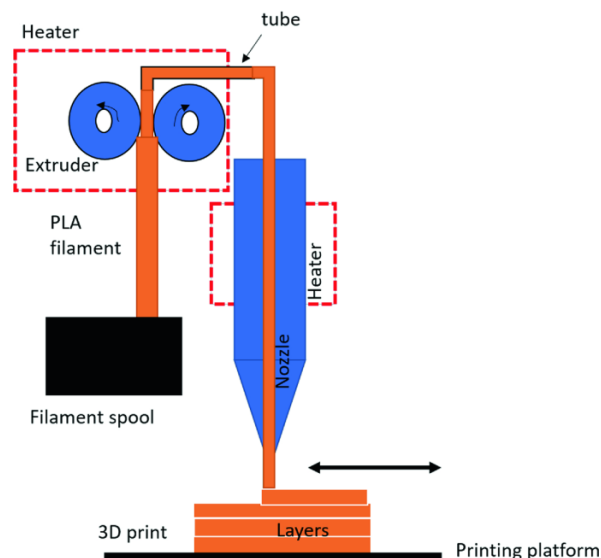
UNIT 3 A

FUSED DEPOSITION MODELING (FDM)

3.1 Fused Deposition Modelling (FDM)

Fused deposition modelling is an additive manufacturing technology that creates 3D components using a continuous thermoplastic or composite material thread in filament form. An extruder feeds the plastic filament through an extruding nozzle, which is melted and then selectively deposited layer by layer onto the build platform in a predetermined automated path.

FDM, also known as Fused Filament Fabrication (FFF), FDM is the most widely used 3D printing technique, with the most 3D printer users globally, and is typically the first 3D printing technology to which people are exposed. Scott Crump pioneered the process in the 1980s under the registered term fused deposition modelling (FDM). Stratasys Inc, a business co-founded by Scott Crump, owns the trademark fused deposition modelling (FDM) and its abbreviation FDM.



3.2 FDM Process:

FDM process is explained in following steps.

The first few steps are similar to any other additive manufacturing technology, starting with build preparation software.

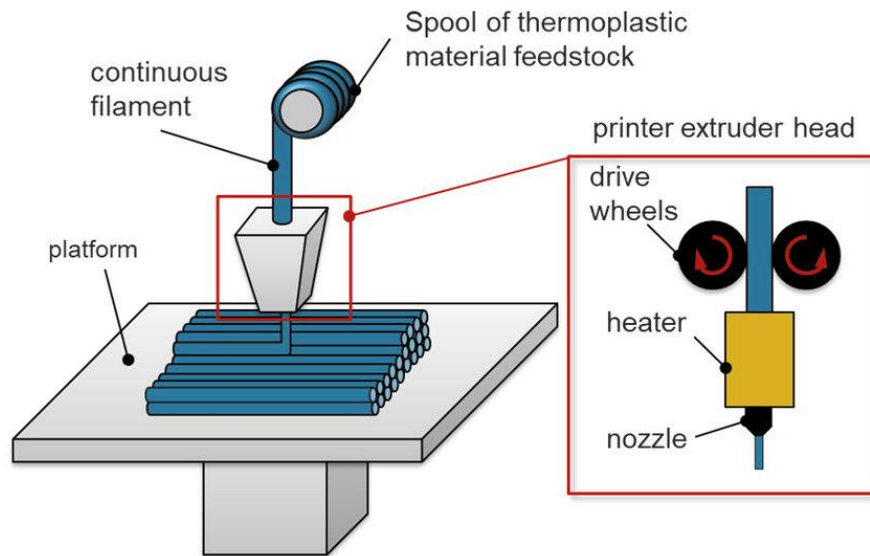


Fig: Fused deposition modelling

1. CAD MODEL DESIGN.

2. Conversion to STL

3. File transfer to slicing software: The software computes sections and slices the part into several layers.

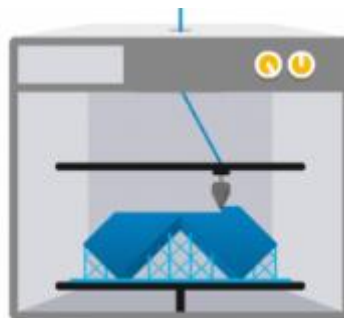
- Slicing software is a category of 3D printing software that is used to convert a basic 3D computer object into something that the printer can understand and print accurately.
- There are various options of slicing software but they all work with the same output.
- They can take a 3D object, convert the surface into miniature triangles, that come together to make the object.
- The amount of these triangles also determines the accuracy and detail in the 3D object that can be printed.
- After setting the desired parameters, the slicer software can then slice the 3D object into the desired layer height and visualized the printing process.
- When this is completed, the object file is then converted into G-code, which a data type used in many manufacturing processes that stores information about how to print
- This G-code is what drives the extruders in the 3D printer to accurately create the object.

4. Transfer to the printing Machine: import the designed and sliced file to the 3D printing machine. and then choose options for the build, such as layer height, orientation

and infill percentage. The program then creates extruder paths and building instructions based on the sectioning data to drive the extrusion heads.

Depending on the printer and the manufacturer, the above process will be different, but the core step of 3D file conversion into layer-based information is the same.

5. **Machine setup:** Then printer is loaded with a thermoplastic filament spool for both model and support extruders. Generally, the build platform is heated and maintained at a higher temperature to control the cooling of the extruded material. Extruders are heated and when the nozzle reaches the required temperature, the head will start pushing and melting the filament into a small ribbon roughly the size of a human hair.
6. **Build:** The extrusion head and the build platform are on a three-axis system, which allows the nozzle tip to move in three directions in space. The extruder will start depositing the material layer by layer in predefined areas to cool and solidify. Sometimes the material cooling is assisted using cooling fans mounted to the extrusion head. Multiple passes are necessary to fill a region within a layer. When the gantry completes a layer, the build platform or the heads will move the Z-axis by the layer height. Then the above process starts again to deposit a new later. This procedure continues until all the layers are built.



7. **Part Removal:** Like any other 3D printing process, the next stage involves removing parts from the build platform and cleaning them by removing all supports.
8. **Post-processing:** Part can then be further processed, remove any remaining supports and finished to suit the end application.

3.3 Characteristics of FDM

When using FDM as the manufacturing process, a designer should keep its capabilities and limitations in mind to produce the best results.

Temperature and build speed: The nozzle and build platform temperature, build speed, layer height and cooling fan speed are all adjustable in most FDM systems. These are often set by the printing service provider and varies with the material.

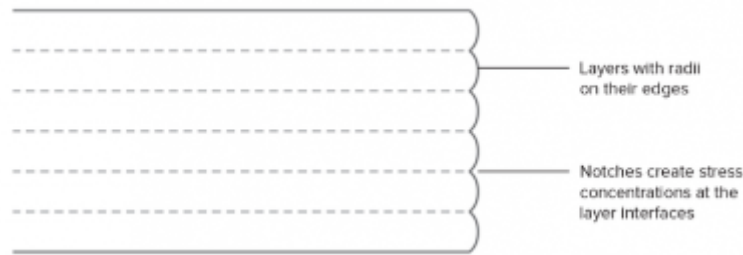
Build volume: Build volume is the biggest part the machine can build. A DIY 3D printer's build volume is typically 200 x 200 x 200 mm, while industrial machines can have build volumes as large as 1000 x 1000 x 1000 mm. Consider the build volume of the printer you will be using during the design.

Layer height: The layer height used in FDM ranges between 0.02 mm and 0.4 mm. Reduced layer height generates smoother components and more correctly captures curved geometries, but a larger layer height make parts print faster and at a lesser cost. Generally, a layer height of 0.2 mm is a good compromise between cost, time and quality. For low fidelity rapid prototypes, the increased layer height expedites the process.

Layer Adhesion: For an FDM component, good adhesion between the deposited layers is critical. The molten thermoplastic is forced against the preceding layer as the nozzle extrudes the current. The high temperature and pressure re-melt the surface of the previous layer, allowing the new layer to connect with the previously printed portion. The binding strength between the multiple layers is always less than the material's base strength.

Hence FDM components are anisotropic by nature and the strength in the Z-axis is always less than their strength in the XY plane. As a result, while designing components for FDM, it is critical to keep part orientation in mind, especially for functional parts.

For example, the tensile strength of ABS with 50% infill in the X and Y direction is approximately four times more than in the Z direction. It also stretched about ten times more before breaking.



Layer by layer-construction FDM (Source:hubs.com)

Furthermore, because the molten material is forced against the preceding layer, it deforms to an oval shape. Hence FDM items will always have a wavy surface, even at low layer height, and that minor features like small holes or threads may require post-processing after printing.

Support Structure: In FDM, geometries with overhangs will require a support structure. The molten thermoplastic cannot be deposited in the absence of air. As a result, some geometries necessitate support structures.

Surfaces printed on supports will have lesser surface quality than the remainder of the item. As a result, it is advised that the part be constructed in such a way that the need for assistance is minimized.

Typically, support is printed using the same material as the print. In industrial printers, there are other support materials available where it can dissolve in liquid, however, they are mostly utilized in high-end desktop or industrial FDM 3D printers. Printing on dissolvable supports enhances the surface quality of the item greatly but raises the total cost of a print due to the need for a dual head FDM printer and the comparatively high cost of the dissolvable material.

Infill and Shell thickness: FDM pieces are typically not produced solid to save time and material. Instead, the exterior perimeter, known as the shell, is printed using numerous passes, and the inside with an internal, low-density structure, known as the infill.

The infill and shell thickness of a print influences the strength of a component.

The default option for desktop FDM printers is 25% infill density and 1 mm shell thickness, which is a fair balance between strength and speed for rapid prints.

- Slicing software gives you the option to print a fully solid object, or a hollow one and everything in-between.
- This is usually how the internal structure of an object is to be printed and the settings range from 5% infill to 100%.
- Furthermore, you get four main infill pattern types which are honeycomb, wiggle, rectangle and triangular.
- Honeycomb offers the greatest strength with minimum material, triangle offers better lateral load strength, which gives the outer shell of the object better strength to handle horizontal forces applied to the object.
- Wiggle is mainly used for flexible materials and rectangular doesn't have any specific advantages.



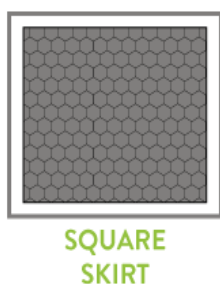
Skirts and Brims

Brims, skirts and rafts are parts that are used to create better bed adhesion for the 3D printed object before the printing process.

A skirt is an outline of the print area of the object, but doesn't directly touch the object.

A brim is directly attached to the object, but goes a little further out and involves more outlines than the skirt.

Its main purpose is to hold the object and to ensure the first layer is printed correctly.

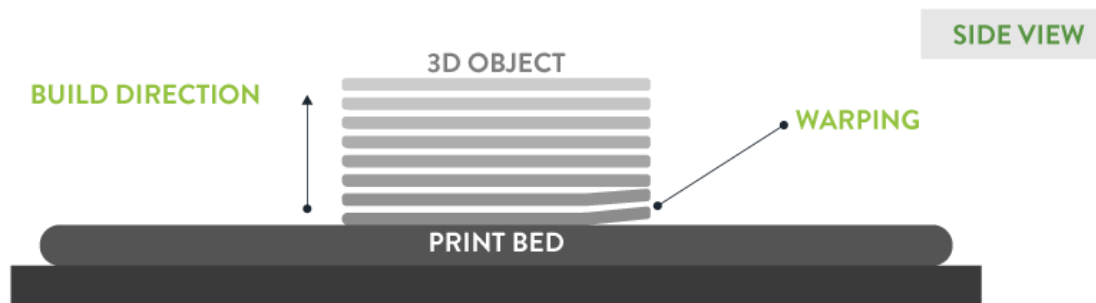


Common Troubleshooting Terms

Warping

Warping is one of the most common FDM flaws. The dimensions of the extruded material shrink as it cools during solidification. Because various parts of the print cool at different rates, their dimensions alter at varying rates as well. Differential cooling causes internal tensions to build up, pulling the bottom layer higher and causing it to distort.

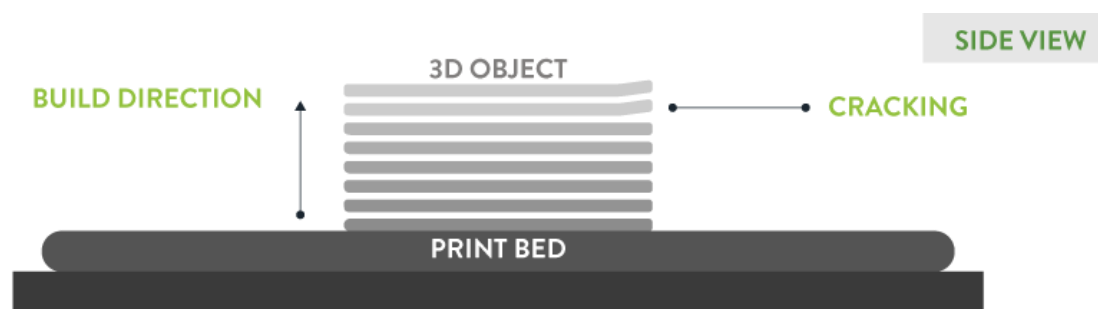
Warping may be avoided by closely monitoring and controlling the temperature of the chamber and the build platform. Good adhesion between the component and the build platform also would help reduce the warping.



- Warping can be described as the shrinkage of a 3D printed object at the corners of the base, mainly attributed to temperature changes.
- Warping is due to the process of non-uniform cooling where certain printing layers cool faster than the heated parts.
- When this occurs, the cooler layers end up distorting the objects geometry since cooling causes shrinkage and this action affects the immediate molten layers.
- As the areas cool and harden, they pull on other layers as cooling increases.
- The main reason for warping is heated thermoplastics need uniformed cooling after being extruded to allow an object to accurately settle while maintaining the desired geometry.
- If the printing bed is not heated or the ambient temperature of the print chamber is not modulated then this leads to different cooling rates.
- To prevent warping issues, ensure that the 3d printer has a heated bed with a metal plate.

Cracking

- Cracking is caused by the same issue as warping, being the non-uniform cooling of a printed object.
- The difference of cracking from warping is cracking occurs at different locations in the printed object.
- To offset the possibility of cracking, a printer with an enclosure allows to eliminate the ambient temperature fluctuations that may occur that could lead to cracking of a 3D printed object.



Fused deposition modelling materials

The variety of materials accessible is one of FDM's primary features. Commodity thermoplastics such as PLA and ABS to engineering materials such as PA, TPU, and PETG. For prototyping, during embodiment stages of the new product design, there are high-performance thermoplastics such as PEEK (polyetheretherketone) and PEI (Polyetherimide).

ABS (Acrylonitrile butadiene styrene)

ABS-M30 is an excellent material for conceptual modelling, functional prototyping, production tools, and end-use parts. ABS-M30, which is up to 70% stronger than conventional FDM ABS, is perfect for production components, thermoforming tools, lightweight jigs and fixtures, and concept models. This thermoplastic has high tensile, impact, and flexural strength. Sparse or solid fill is available.

- Pros – Offers good strength, good temperature resistance
- Cons – More susceptible to warping

PC

PC (polycarbonate) is widely used in the automotive, aerospace, and medical industries, among many others. PC provides precision, durability, and stability, resulting in sturdy parts that can resist functional testing. Rapid tooling, jigs and fixtures for production

- Pros – Accurate, Rigid, Stable, RF transparent, High tensile and flexural
- Cons – Not widely available, higher cost

PC-ABS

PC-ABS combines the best qualities of both PC and ABS materials, such as strong strength with the heat resistance of PC and the flexibility of ABS. PC-ABS composites are widely used in automotive, electronics, and telecommunications applications. Form fit and functional prototypes, low volume production parts

- Pros – High impact strength, High heat resistance
- Cons -Not widely available, higher cost

PLA

PLA is a bioplastic (Biodegradable) and is one of the standard materials for this technology, along with ABS (Acrylonitrile Butadiene Styrene).

- Pros – Excellent visual quality, Easy to print with
- Cons- Low impact strength

Nylon

Nylon is one of the most widely used commercial FDM materials and it possesses a good combination of tensile strength and toughness. Generally used for rapid prototyping concept validation models, prototype parts for visual design validation, production toolings such as jigs, fixtures, and manufacturing aids. It can also be used for low volume production parts with less demanding functional parts.

- Pros – High strength, Excellent wear, and chemical resistance
- Cons – Low humidity resistance

PETG (Polyethylene terephthalate glycol) Polyethylene terephthalate glycol, commonly known as PETG or PET-G, is a thermoplastic polyester that provides significant chemical resistance, durability, and excellent formability for manufacturing. PETG can be easily vacuumed and pressure-formed as well as heat-bent thanks to its low forming temperatures. It is mainly use in Food and Drink Containers, Medical and Pharmaceutical Applications and etc.

- Pros – Food safe although grooves and grannies of notches between the layers are a critical point for bacterial growth, good strength, easy to print
- Cons – Not widely available, higher cost

TPU (Thermoplastic polyurethane)

TPU 92A FDM Elastomer is a thermoplastic polyurethane substance used to make long-lasting elastomer components. The material allows for the development of high-functioning, long-lasting, and complicated parts with the anticipated material features of an elastomeric material, such as enhanced tear resistance, fatigue resistance, and memory recovery.

- Pros – Very flexible, good toughness, Durable, Abrasion resistance
- Cons – Difficult to print accurately

PEI (Polyetherimide) Characterized by a combination of outstanding thermal, mechanical and electrical properties, Polyetherimide (PEI) has made its place in high performance applications like automotive, aerospace, industrial and many more.

- Pros – Excellent strength to weight ratio, Excellent fire and chemical resistance
- Cons – High cost

Applications of FDM

What may FDM 3D Printing be used for? Although the options for product creation and manufacture are limitless, the majority of applications fall into four broad categories:

- Functional prototypes
- Production and manufacturing tools
- Concept models
- Production quality parts

Advantages of FDM

- FDM is the most cost-effective method of manufacturing bespoke thermoplastic components and prototypes.
- Due to the lower cost of FDM printers and wide availability, the lead times are minimal and cheaper than other additive manufacturing processes.
- There is a large variety of thermoplastic materials available for prototyping as well as certain non-commercial practical. Most common injection moulding materials can be replicated using FDM
- The technology is aesthetically pleasing, easy to use, and appropriate for the workplace.
- Mechanical and environmental stability are provided by supported production-grade thermoplastics.

- FDM technology makes complex shapes and voids that would otherwise be impractical possible.

Disadvantages of FDM

- When compared to other 3D printing methods, FDM has the lowest dimensional accuracy and resolution, making it unsuitable for items with delicate features.
- Because FDM items are prone to noticeable layer lines, post-processing is essential for a smooth finish.
- Because of the layer adhesion technique, FDM components are inherently anisotropic.